A first split away from the anchor

Human Compression project: a component of the L019 recovery

We retain the actual critical configuration of the L019 recovery. The full marking is a fractured natural symmetric string C-marking on $X \cup Y$, where $|X| = m$ is even, $|Y| = 6$, and $X \cap Y = \{e\}$. Its labels are the concatenation of an even degree-six rank-three A_5 factor E on Y and a natural symmetric factor

$$P = \langle t_0, \dots, t_{s-1} \rangle = \mathrm{Sym}(X), \qquad s = m/2 + 1.$$

The common point e is an anchor at the last mark of E and the first mark of P. Both factors are fractured. The full degree and rank are $n = m + 5$ and $r = m/2 + 4$, so $2r = n + 3$.

An internal perfect split is called *sufficient* if its split mark is pure, or if one internal factor is trivial and the two actual groups meeting in the crossing point both contain an odd permutation. The acquired pure-split and odd-attachment theorems give the required subdivision in these cases. In particular, when the split mark is $q = (c\ d)\beta$ and its supported wings are F, R, these two groups are F and $\langle q, R \rangle$, not merely F, R.

Assume throughout the strict smaller-degree sufficient-split induction: every fractured natural symmetric string C-marking of degree $d < n$ and rank u with $2u > d + 2$ has an internal sufficient split. No critical factor at equality is assumed solved by this hypothesis.

Lemma 1. *Let j be the least colour of P with exactly one eligible edge. It is a perfect split. Write A, B for its omission orbits, with $e \in A$, and put $a = |A|$, $b = |B|$. Then $2j \le a - 1$, the right wing is the actual* $\mathrm{Sym}(B)$*, and $j \neq 1$.*

Proof. The acquired anchored first-split theorem applies to P: it is transitive and fractured, has the given anchor, and $2s = m + 2 > m$. It supplies perfection and the stated density. The right wing has rank $q = s - j - 1 = m/2 - j$, whence

$$2q = b + a - 2j \ge b + 1. \tag{1}$$

Use the precise nonsymmetric-wing conclusion of the acquired actual-wing theorem. It bounds each actual nonsymmetric wing by half its degree, independently of the stronger parent-rank inequality used only in that theorem's final either-wing conclusion. Its full L047 exception list and its A_{11} attachment obstruction are retained inputs here. Equation (1) therefore forces the right wing to be $\mathrm{Sym}(B)$. If $j = 1$, its single left involution would be transitive on A, so $a = 2$, contrary to $2j \le a - 1$. □

Lemma 2. *An internal sufficient split in an actual symmetric wing with $2u > d+2$ transfers to an internal sufficient split of the full marking. The same holds for a symmetric factor supported on one of two sets meeting in one point, with consecutive label intervals and the stated endpoint anchors.*

Proof. For a wing use the acquired internal split-inheritance proof. An internal index has distance at least two from the old split mark. That mark commutes with the relevant transitive side and with the new unique crossing, and hence fixes that side pointwise. The opposite side becomes its old side together with the other wing, and remains transitive. The new crossing and its internal factors are unchanged. Purity and a trivial factor therefore persist. Each new

attaching group contains its corresponding old attaching group, so each old odd element persists as an actual permutation. This also preserves sufficiency.

For factors meeting in one point, use the acquired one-point transport lemma: the side containing the shared anchor is enlarged by the other factor, and the other side is fixed. The same containment and parity argument applies. All such submarkings are fractured, since a transitive omission there would become a transitive omission of the full marking. Their restrictions to their actual supports are faithful and inherit IP; no quotient is used. □

Theorem 3. *Under the standing induction hypothesis, if $j > 0$ in Lemma 1, the full marking has an internal sufficient split. Consequently, a remaining critical configuration may be assumed to have $j = 0$.*

Proof. By Lemma 1, $j \geq 2$ and $a \geq 2j + 1 \geq 5$. The split of P is also a perfect split of the full marking. Its left wing

$$F = \langle E, t_0, \ldots, t_{j-1} \rangle$$

is supported and transitive on $Y \cup A$, of degree $a + 5$ and rank $j + 3$. Indeed E and the prefix are transitive on sets meeting at e. The right wing $R = \mathrm{Sym}(B)$ is unchanged. The split mark t_j fixes Y pointwise, and has the form

$$t_j = (c\ d)\alpha\beta, \qquad c \in A, \quad d \in B,$$

with the two internal factors on their respective sides. Set $\delta = a - 2j \geq 1$. Equation (1) becomes

$$2q = b + \delta. \tag{2}$$

The right wing is nontrivial: otherwise $b = 1, q = 0$ would contradict (2).

Suppose first that both internal factors are nontrivial. Apply the acquired faithful-augmentation lemma with R as its symmetric wing, reversing the labels. The other augmented factor has a faithful primitive C-action K of degree $a + 6$ and rank $j + 4$. That lemma excludes both natural alternating and symmetric groups and supplies its nontrivial first-mark character. If $\delta \leq 2$,

$$2(j + 4) = (a + 6) + 2 - \delta \geq a + 6.$$

The complete primitive theorem of L047 now applies at or above half the degree. Its proper exceptions have degrees six and ten, while $a + 6 \geq 11$. Thus this case is impossible. It follows that $\delta \geq 3$. Equation (2) gives $2q > b + 2$, so induction in the smaller actual symmetric wing R, followed by Lemma 2, proves the assertion.

Suppose next that $\alpha = 1$. Since $\beta \in R$, the group $J = \langle t_j, R \rangle$ is the actual $\mathrm{Sym}(B \cup \{c\})$. It is supported on that set, meeting the support of F only at c. If F is odd, both attaching groups are odd and this split is sufficient. If F is even and $\delta \geq 2$, then

$$2\,\mathrm{rank}(J) = b + \delta + 2 \geq |B \cup \{c\}| + 3.$$

Induction in J and one-point transport apply, since its degree $b + 1 < n$.

It remains in this case that F is even and $\delta = 1$. Then its degree $f = a + 5$ is even, $f \geq 10$, and its rank is $f/2$. It is fractured and has an endpoint anchor at c. The acquired anchored-even theorem excludes $f \geq 12$. At $f = 10$, the acquired small-even theorem applies with the actual odd external group J. Its only permissible even half-degree factor is the degree-six A_5, so degree ten is excluded too. These invocations retain the exact actual attachment and supported-kernel hypotheses supplied by F, J.

Finally suppose $\beta = 1$; the case where both factors are trivial already gives a pure split. Reverse the split. The two factors meeting at d are now R and $H = \langle F, t_j \rangle$, supported on B and $Y \cup A \cup \{d\}$. The symmetric group R is odd. If H is odd the split is sufficient. If H is even, the acquired even-attachment theorem says either that $2q > b + 2$, or that H is the degree-six

A_5. The latter is impossible since $\deg(H) = a + 6 \geq 11$. In the former case induction in R and wing transport finish the proof.

Every induction above uses strictly fewer than n points and the strict rank inequality. If a resulting degree were at most four, that strict inequality is already impossible: a fractured transitive family on d points has at most $d - 1$ members by the acquired one-edge orbit forest. Thus no unstated small base or critical equality is being invoked. □

Attribution and remaining case. The rank classifications, actual-wing theorem, faithful augmentation, even-attachment and finite even-factor exceptions remain fully attributed in the acquired components. The present deduction combines them with the new first-cut density. It does not make an arbitrary central quotient into a C-group. If $j = 0$, the symmetric factor begins with its unique crossing at its anchor, but that mark may still have a nontrivial internal factor. The remaining endpoint configuration and the stable counting correspondence remain unproved here. No whole-original, conjecture, Lean or measurement credit is asserted.